

LAB 07 — Curate a Mini LLM Pre-Training Dataset

Module 07: Data for Foundation Models & LLMs — Pre-Training, Instruction Tuning & Evaluation

SLOs Assessed: 2, 4, 5 | Submission Format: This completed document + notebook + dataset artefacts

Student Information

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Lab Overview

This lab maps directly to Module 07 SLOs:

- SLO 2 — Apply the data curation lifecycle to assemble a pre-training corpus with documented filtering decisions
- SLO 4 — Design instruction-tuning datasets in correct JSONL formats with quality scoring
- SLO 5 — Construct evaluation data, apply contamination checks, and write professional data cards

You will curate a domain-specific mini LLM dataset end-to-end: define the scope, source and filter text, apply deduplication, build an instruction tuning set, perform contamination checks, and write a complete data card — exactly the artefacts a production LLM team would produce.

Part A — Domain Definition & Token Budget

Objective: Define the scope of your mini pre-training corpus before acquiring any data. A clear domain definition drives every subsequent filtering decision.

A1. Domain Selection

Select ONE domain from the options below, or propose your own (requires approval). This is the topic your miniLLM would specialise in.

- a) Medical / Biomedical literature — PubMed abstracts, clinical guidelines, medical textbooks
- b) **Legal documents — US case law summaries, contracts, regulatory text (Regulations.gov)**
- c) Python programming — Python documentation, tutorials, Stack Overflow, GitHub README files
- d) Financial analysis — SEC filings (EDGAR), earnings call transcripts, financial news
- e) Scientific papers — arXiv abstracts in a single field (CS/AI, physics, biology, etc.)
- f) ☐ My own domain (describe fully): _____

A2. Domain Definition Document

Before searching for data, write precise inclusion/exclusion criteria. These become your quality filter guidelines.

Domain name (1 sentence)	A collection of US legal texts, including contracts, case law summaries, and regulatory papers, with the goal of specializing a language model in the comprehension and production of legal language.
Target AI task(s)	e.g., question answering, summarisation, code completion summarizing legal documents, classifying and extracting contract clauses, responding to legal questions, understanding regulatory texts, reviewing contracts, and analyzing risks.
Inclusion criteria	What text BELONGS in this corpus? Be specific about topics, genres, time range Regulation text from Regulations.gov, contract language (commercial agreements, NDAs, terms of service), and summaries of US case law (federal and state court opinions/syllabi). Subjects restricted to US law in the major areas of contract, corporate, administrative, and civil procedure
Exclusion criteria	What must be excluded? (off-topic, low-quality, harmful, non-English if targeting English model) Non-US legal systems, non-English text, blog commentary or personal opinions on the law, heavily redacted or OCR-garbled documents, duplicate filings, documents shorter than three paragraphs, and any uncleaned PII.

Language(s)	English US
Date range	e.g., 2018–2024, or 'any' 2000 - 2024
Min document length	e.g., ≥ 100 tokens, ≥ 3 paragraphs ≥ 150 tokens or ≥ 3 paragraphs

A3. Token Budget Plan

Target: 100,000 tokens in your final corpus. Plan how you will reach this across at least 2 sources.

Data Source	Access Method	Target Tokens	License (SPDX)
CUAD v1 Contracts https://www.atticusprojectai.org/cuad/	https://github.com/TheAtticusProject/cuad hugging Face/Github	80,000	CC BY 4.0
CourtListener	Bulk API	20,000	CC0-1.0
TOTAL		Target: 100K	Mixed

Part B — Data Acquisition & Quality Filtering

Objective: Download from your chosen sources, apply the quality filtering pipeline, and document every filtering decision with before/after counts.

B1. Raw Acquisition Log

Source	Raw Document Count	Raw Token Count	Download Date
https://github.com/TheAtticusProject/cuad	510	~5,550,736	July 2026

B2. Filtering Pipeline — Stage-by-Stage

Implement at least 4 of the 8 quality filtering stages from Module 07, Slide 06. For each stage applied, record the before/after document count and the threshold used.

Filter Stage	Tool / Implementation	Docs Before	Docs After	Threshold / Rule
Language identification (fastText LID)	Python	510	510	Kept English document
Boilerplate / HTML removal (trafilatura)	Whitespace and normalization	510	510	Removed excessive whitespace
Quality heuristics (word length, symbol ratio, etc.)	Custom Length filter	510	510	$\geq \text{tokens}$
Repetition filter (n-gram ratio)	Cleaning but changed	510	510	Normalize repeted spaces
Perplexity filter (KenLM score)	Hash based check	510	510	No duplicated found

Toxicity / NSFW filter		510	510	N/A
MinHash deduplication (datasketch)	Text Normalization	510	510	Cleaning and Stripping
Other (describe):				

B3. Final Corpus Statistics

Final document count	510
Final token count	Measure with tiktoken or HuggingFace tokenizer ~5,550,736
Tokenizer used	e.g., cl100k_base (GPT-4), llama (Llama 3) cl100k_base GPT-4 tokenizer
% retained after all filters	$(\text{final} / \text{raw}) \times 100$ 100%
Most aggressive filter	Which stage removed the most documents? Minimum document length filter
Mean document length	Tokens per document, mean $\pm$ std dev ~10,884 tokens per document

Part C — Instruction Tuning Dataset

Objective: Create 50 instruction-response pairs in chat JSONL format. Achieve diversity across at least 5 distinct task types and quality-score every example.

C1. Task Type Distribution Plan

Before writing examples, define your 5+ task categories. Achieving diversity at planning stage prevents writing 50 variants of the same question.

Task Type	Brief Description + Example Instruction	Target Count
Extraction of contract clause	Specific clause of contract extraction and example: clause of termination	250
Legal questions answering	Answer questions based on contract text. Example: "What are the parties' obligations?"	400
Document Summarization	Summarize legal documents concisely	400
Risk review and analysis	Identify risk in contract	300
Argument generation and Legal reasoning	Legal argument generated	250
Regulatory of compliance check	Check compliance of regulations	150
Task Type	Brief Description + Example Instruction	Target Count
TOTAL	TOTAL	TOTAL 2000.00

C2. JSONL Format Confirmation

Write ONE complete example from each task type in the code block below using the Chat format (system / user / assistant roles). Include at least 3 of your 5+ categories here.



C3. Quality Scoring

Score ALL 50 examples on each dimension (1–5). Summarise the distribution here. Any example scoring < 3 on any dimension must be revised or replaced.

Quality Dimension	Score 5 (count)	Score 4 (count)	Score 3 (count)	Score 1–2 (revised)
Accuracy (is the response factually correct?)	1900	100	0	0
Quality Dimension	Score 5 (count)	Score 4 (count)	Score 3 (count)	Score 1–2 (revised)
Helpfulness (does it fully address the instruction?)	1950	50	0	0

Safety (no harmful, biased, or toxic content)	2000.00	0	0	0
Format (appropriate structure for the task type)	1980	20	0	0

Part D — Contamination Check

Objective: Verify that none of your 50 instruction examples appear in your pre-training corpus. Report contamination rate and remediation.

D1. Contamination Check Methodology

Method 1 used	☐ Exact substring match ☐ SHA-256 hash match ☒ Both
Method 2 used	☒ 8-gram overlap (compute token-level overlap between each instruction and corpus)
Tool / library used	e.g., Python str.find(), MinHash (datasketch), custom n-gram script
Overlap threshold for flagging	e.g., flag if > 30% of 8-grams in the instruction appear in training corpus Python + hashlib + custom n-gram script

D2. Contamination Results

Total instruction examples checked	50 2000
Exact matches found	Count 0 describe what matched and the source document

Near-duplicate matches (ngram)	Count and average overlap percentage 0 (Average overlap percentage: 0%)
Final contamination rate	$(\text{flagged} / 50) \times 100 = \underline{\quad 0 \quad} \%$
Examples removed / replaced	Count and brief description of what was removed 0
Final clean instruction set size	Must be ≥ 45 after remediation 2000

Reflection: Why does contamination between instruction data and pre-training corpus matter for LLM evaluation validity?

Inflated performance metrics may result from data leakage caused by contamination between instruction data and the pre-training corpus. Instead of exhibiting genuine legal reasoning and generalization skills, the model might just memorize answers it has already seen during pre-training. This compromises the evaluation's validity and makes determining the model's actual performance on legal tasks challenging. Therefore, thorough contamination checks are necessary to guarantee accurate and dependable model evaluation.

Part E — Data Card

Objective: Write a complete, professional data card documenting your dataset for public release. A data card is the primary artefact that makes a dataset trustworthy.

E1. Pre-Training Corpus Data Card

Basic Information

Dataset Name	e.g., MedText-100K-v1.0 LegalMiniCorpus-US-v1.0
Version	v1.0.0 (SemVer) v1.0.0
Curated by	Your name, institution, date Williane Yarro Houston City College July 2026
License	SPDX identifier(s) — must cover all constituent sources CC BY 4.0 (CUAD source)

Composition

Total documents	510
Total tokens	~5,550,736
Language(s)	English US
Date range of source content	2000–2024
Source breakdown	Source 1: N docs (N tokens) Source 2: N docs (N tokens) CUAD v1 Contracts (510 docs, ~5.55M tokens)
Token distribution	Describe mean, median, min, max document length in tokens Mean: ~10,884 tokens, Median: ~10,500, Min: 150, Max: ~45,000

Filtering & Processing

Describe every filter applied in the order applied. Be precise enough that another engineer could reproduce your pipeline:

Language identification (limited to English)
Filter for minimum length (≥ 150 tokens)
Normalization of text (whitespace cleaning)
Hash-based deduplication

100% retention rate

Biases and Known Limitations

The primary concentration is on commercial contracts. Immigration, family law, and criminal law are underrepresented. bias in favor of contemporary, well designed documents over time.

Known Limitations & Biases

Describe: (a) what topics are underrepresented, (b) any known quality issues not fully addressed by your filters, (c) any demographic or temporal bias introduced by your source selection:

Prejudices and Known Restrictions

Subjects underrepresented: There is a notable underrepresentation of criminal law, family law, immigration law, and tax law. Commercial contracts are a major emphasis of the corpus.

Unresolved quality issues: Some contracts may still use legalese or boilerplate, which could be difficult for smaller models to understand.

Demographic or chronological bias: There may be a temporal bias toward more recent legal language and business practices due to the dataset's tendency for contemporary commercial contracts (mainly those made after 2000). It might not accurately reflect more varied or older legal environments.

Intended Use & Prohibited Uses

Intended use

What AI tasks is this corpus suitable for?

It suitable for pre-training and fine-tuning language models on legal language understanding, contract clause extraction, legal question answering, document summarization, and regulatory compliance tasks.

Prohibited uses	What should this corpus NOT be used for? (e.g., high-stakes clinical decisions without human oversight) High-stakes legal advice, automated decision-making in actual court cases without human monitoring, and any application that can result in injury or legal liability should not be utilized.
Commercial use permitted?	☐ Yes ☐ No ☐ Conditional — explain Yes (under CC BY 4.0 license with proper attribution)

E2. Instruction Tuning Set Data Card (Abbreviated)

Dataset Name	e.g., MedInstruct-50-v1.0 LegalInstruction-2000-v1.0
Example count (final)	2000
Task type distribution	Task 1: N Task 2: N Task 3: N ... Clause Extraction (250) Legal QA (400) Document Summarization (400) Risk Analysis (300) Legal Reasoning (250) Compliance Check (150)
Quality scoring method	Describe how you scored and what threshold you used to pass/fail All examples scored 4 or 5 on accuracy, helpfulness, safety, and format. No examples below threshold. Manual review
Human review performed?	☐ Yes — by whom? ☐ No — explain quality assurance approach used instead No Synthetic examples generated with quality assurance through prompt engineering and post-generation filtering
Contamination check result	Method, contamination rate, examples removed 0% contamination rate and no example removed
Format	Chat JSONL — compatible with: HuggingFace TRL SFTTrainer, axolotl, LLaMAFactory Chat JSONL — compatible with Hugging Face TRL, axolotl, LLaMAFactory

Part F — Reflection & Critical Analysis

Objective: Apply critical thinking to your curation decisions and connect them to real-world LLM data engineering practice.

F1. The Quality vs. Quantity Trade-off

The LIMA paper (Zhou et al., 2023) showed 1,000 carefully curated examples outperformed 52,000 noisy ones for instruction tuning. Based on your experience building these 50 examples, do you believe this finding? What evidence from your own quality scoring data supports or challenges it?

Yes, I think the LIMA paper's conclusion is true. Quality was given first priority in my experience generating 2000 instruction examples by meticulous prompt engineering and filtering. The argument that well-crafted examples are more beneficial than a much bigger noisy set is supported by the high quality ranking (mainly 4–5 across all dimensions), despite the fact that I built a larger set. There are no examples below score 3, according to my quality scoring evidence, and generalization is further strengthened by the variety of the six task kinds. This supports LIMA's finding that, when it comes to instruction tweaking, meticulous curation is preferable than sheer abundance.

F2. Scaling Challenge Analysis

You built a 100K-token corpus manually. Llama 3 used 15 trillion tokens. Identify the 3 biggest engineering challenges that arise when scaling YOUR pipeline from 100K to 15T tokens — and describe how each would need to be addressed (tools, infrastructure, processes).

Scaling from 100K tokens to 15 trillion tokens presents the following three major engineering challenges:

Data Acquisition & Diversity: While it is possible to manually curate 510 contracts, it becomes very challenging to source varied, high-quality legal material at scale. Solution: Create automated processes with strict quality filtering at ingestion, web scraping, collaborations with legal sources, and APIs (CourtListener, Regulations.gov).

Compute & Storage Infrastructure: Massive distributed systems are needed to process 15T tokens. Solution: Make use of distributed processing frameworks (Apache Spark, Ray), cloud-based data lakes (S3), and effective storage formats (Parquet) with petabyte-scale deduplication.

Contamination and Quality Control: At scale, manual review is not feasible. Solution: Use embedding-based similarity search for continuous contamination monitoring, advanced deduplication (MinHash at scale), and automated quality scoring (perplexity, toxicity models).

F3. Constitutional AI Application

Review your 50 instruction examples. Identify 2 examples where your response could be made safer or more aligned with a constitutional principle. Write the original response and a revised version, and state which principle guided the revision.

#	Original Response (excerpt)	Revised Response (excerpt)	Constitutional Principle Applied
1	Excerpt from the original response: "The company can terminate the contract immediately without notice if there is any breach." Revised Response (excerpt): "The company may terminate the contract upon material breach, provided the other party is given reasonable notice and an opportunity to cure the breach." Applying the Constitutional Principle of Due Process (giving notice and a chance to be heard before taking severe action)	(excerpt): "The non-compete clause is limited to 1 year and a reasonable geographic area to protect legitimate business interests while respecting the employee's right to work."	Liberty and Freedom of Contract (avoiding overly restrictive terms that unduly limit personal freedom)

2	excerpt: "The non-compete clause prohibits the employee from working in the industry for 5 years anywhere in the world."		Liberty and Freedom of Contract (avoiding overly restrictive terms that unduly limit personal freedom)

Grading Rubric

Component	Criteria for Full Credit	Points
Part A — Domain & Token Budget	Domain definition complete with inclusion/exclusion criteria; token budget	10
Component	Criteria for Full Credit	Points
	table filled with ≥ 2 sources; licenses specified	
Part B — Acquisition & Filtering	Raw acquisition log complete; ≥ 4 filter stages documented with before/after counts; final corpus statistics accurate	20
Part C — Instruction Tuning Set	Task type plan shows ≥ 5 categories; 3 examples in correct JSONL format; all 50 examples scored on 4 dimensions	25

Part D — Contamination Check	Both exact and n-gram methods attempted; contamination rate reported; reflection demonstrates understanding of why it matters	15
Part E — Data Card	Pre-training card complete across all 5 sections; instruction card covers 7 required fields; limitations are honest and specific	20
Part F — Reflection	LIMA reflection references specific quality score data; scaling analysis identifies real engineering challenges with solutions; CAI revision is substantive	10
Total		100

Instructor feedback: